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**Doping Iron in CsPbBr₃ Perovskite Nanocrystals for Efficient and Product
Selective CO₂ Reduction**

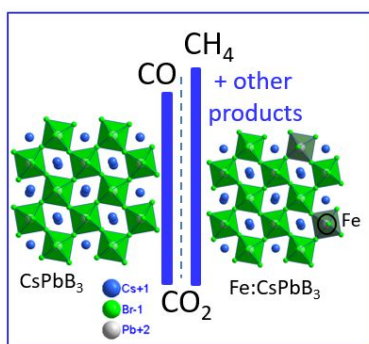
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Abstract: Lead halide perovskite nanocrystals are recently emerged as an efficient optical material for light harvesting. While these were extensively studied for obtaining bright emissions; but little had been explored as catalysts for enhancing the rate of chemical reactions. Considering their importance in catalysis, herein, Fe(II) doped CsPbBr₃ perovskite nanocrystals were explored for photocatalytic reduction of CO₂. In comparison to undoped CsPbBr₃, doped nanocrystals showed enhanced catalytic activity and also predominantly led to evolution of CH₄ than CO. The observation of reverse trend of predominated CH₄ evolution in doped nanocrystals rather than CO observed for undoped nanocrystals was correlated to the adsorption/desorption energy of respective products established theoretically earlier. This selective evolution of major products on doping remained unique and also a step forward for understanding more regarding light to chemical energy conversions using perovskite nanocrystals.

TOC:



Perovskite nanocrystals have emerged as one of the most efficient light harvesting materials in current research.¹⁻⁷ For light to light energy conversion, these have been extensively studied and as high as near unity photoluminescence quantum yields in different colors were established.^{5, 8-14} However, being efficient light absorbing material, their utilizations in light to chemical energy transfer are limited.¹⁵⁻¹⁷ Major obstacles of these nanocrystals are their sensitivity towards polar environments and the strong recombination possibility of the photogenerated charge carriers.^{3, 18, 19} Hence, unlike chalcogenide quantum dots, implementations of these nanocrystals in catalysis is less explored. From the literature reports, it is revealed that, methylammonium lead iodide (MAPbI₃) was used for the first time as a photocatalyst for hydrogen (H₂) generation from HI solution.¹⁵ Soon after several groups extensively studied stable CsPbBr₃ perovskite nanostructures for photocatalytic CO₂ reduction.^{16, 19-21} Creating heterojunctions with highly conductive material (e.g. CsPbBr₃/graphene QD, g-C₃N₄, Pd nanosheet), carrier recombination possibilities were reduced and efficiency of the catalytic activities of these materials were enhanced.^{16, 19, 20, 22} Again, tuning the size and composition of the CsPbBr₃ perovskite nanocrystals, the productivity was also greatly enhanced.²¹ Beyond the reduction, controlling the selectivity of products from CO₂ reduction is also important and using CsPbBr₃ nanocrystals as photocatalysts, this aspect of controlling the products is also limited.¹⁹

Recently, Tang et al. theoretically predicted the adsorption energies of different products in various metal doped CsPbBr₃.²³ Inspired from this report where Co and Fe were proposed as ideal dopants to predominantly produce methane, herein, Fe(II) doped CsPbBr₃ nanocrystals were synthesized and explored for photocatalytic reduction of CO₂. With 3% Fe(II) doping (with respect to Pb), the obtained doped CsPbBr₃ showed selectively CH₄ as the dominated products where the undoped CsPbBr₃ led to CO as the major product. In addition, the catalytic performance for the doped nanocrystals was also observed enhanced. The efficient photocatalysis is also supported with measuring the transient current and comparing the catalytic activities in several controlled reactions. Post catalysis samples were also vividly characterized and the presence of Fe as the key for twisting the catalytic path for these nanocrystal catalysts was established. To our best knowledge this is a unique report where catalysts were used without any additional carrier transportation material or hetero-structures.

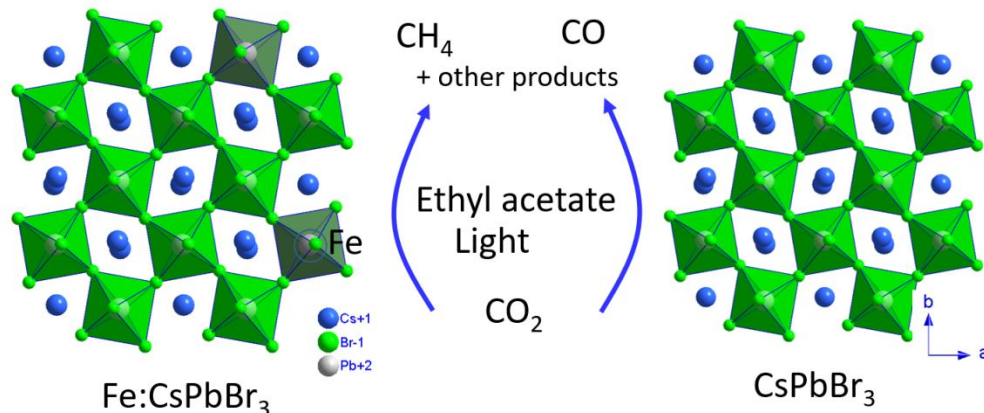


Figure 1. Schematic presentation of Fe doped CsPbBr₃ and undoped CsPbBr₃ nanocrystal photocatalysts leading to CH₄ and CO as the major product respectively in photocatalytic CO₂ reduction reactions.

For the synthesis of CsPbBr₃ nanocrystals, reported literature method¹ was adopted and for doping, an optimized 25% of Fe(II) precursor (with respect to Pb) was introduced into the reaction system before injection of Cs(I) precursor. The doping of Fe(II) was expected to replace of Fe(II) in the position of Pb(II) in the lattice. Atomic models for the doped and undoped CsPbBr₃ nanocrystals are shown in Figure 1. The selectivity of products CH₄ and CO from doped and undoped nanocrystals which were experimentally found and discussed latter, also schematically shown in this figure.

Figure 2a shows the absorption and photoluminescence (PL) spectra of as synthesized hexane dispersed undoped and doped CsPbBr₃ nanocrystals. Their spectral nature and positions remained similar and as expected from reported CsPbBr₃ nanocrystals.^{1,3} Figure 2b presents the Transmission electron microscopic (TEM) image of undoped CsPbBr₃ nanocrystals and Figure 2c shows TEM image of Fe(II) doped nanocrystals. Size of these nanocrystals remained 12-15 nm and dispersity also retained identical. However, with higher amount of Fe(II), the disparity remained wider (Figure S1) and also the catalytic activities reduced (discussed later section).

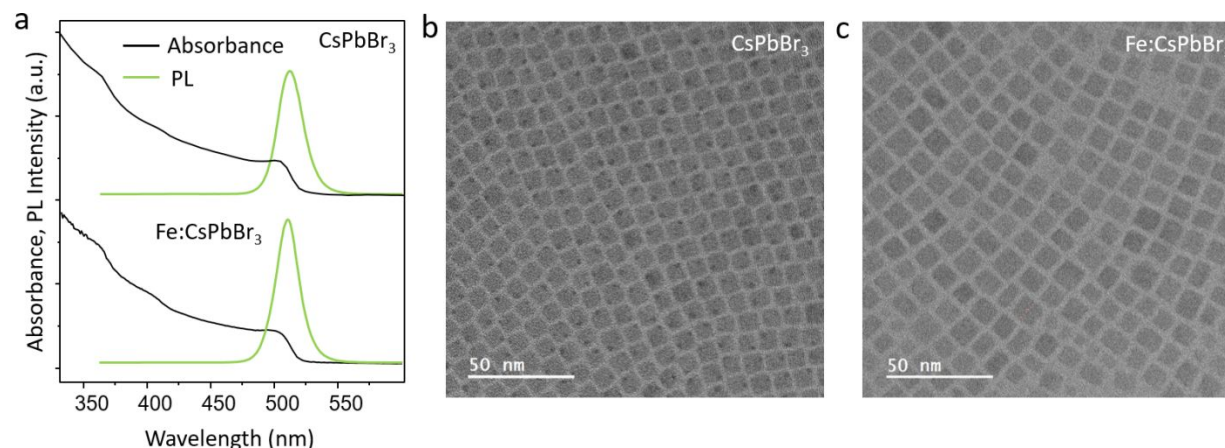


Figure 2. (a) Absorption and photoluminescence (PL) spectra of as synthesized CsPbBr₃ and Fe(II) doped CsPbBr₃ nanocrystal photocatalysts. Excitation wavelength was 350 nm. TEM images of (b) CsPbBr₃ and (c) Fe(II) doped CsPbBr₃ nanocrystals.

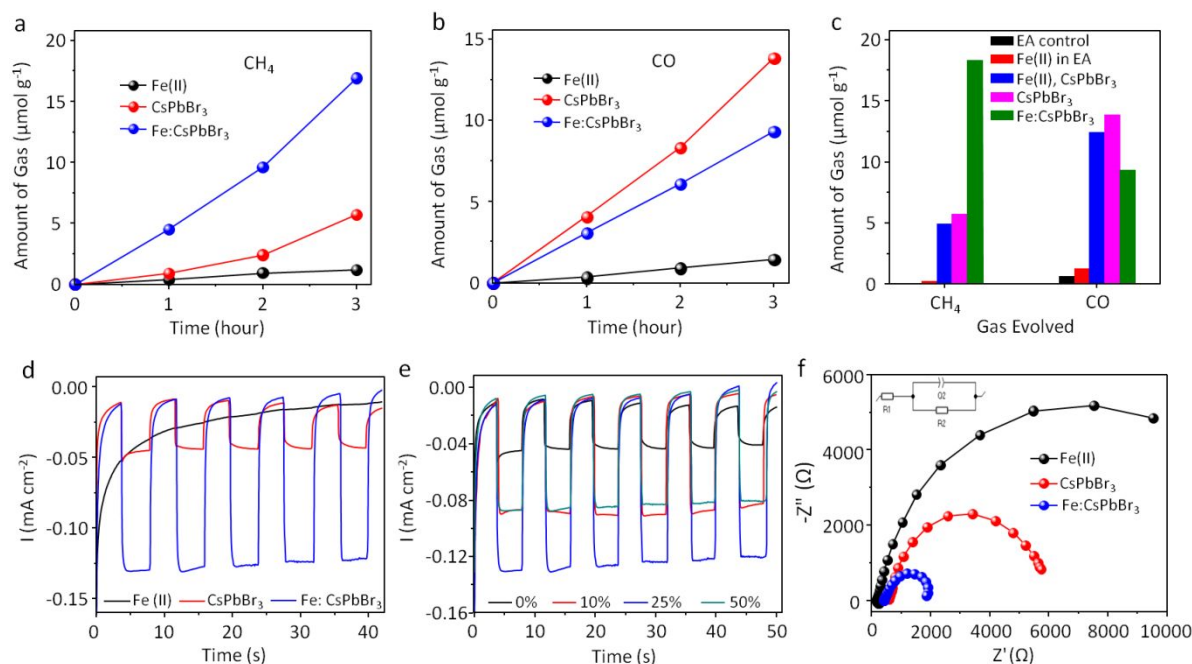


Figure 3. Quantitative evolutions of (a) CH₄ gas and (b) CO gas from CsPbBr₃, Fe(II) doped CsPbBr₃ nanocrystals as photocatalysts and the control reaction using only Fe(II) acetate salt during the process of photocatalytic reduction of CO₂ with the course of time. (c) Bar diagram showing relative ratios of CH₄ and CO evolved using Fe(II)acetate addition to CsPbBr₃, CsPbBr₃, Fe(II) doped CsPbBr₃, Only Fe(II)acetate without CsPbBr₃ and the control reaction without any additive.

(d) Transient photocurrent response of photoelectrodes having CsPbBr₃, Fe(II) doped CsPbBr₃ nanocrystals and the control reaction using only Fe(II)acetate salt. (e) Transient photocurrent response of photoelectrodes having doped CsPbBr₃ prepared with different Fe(II) loading with respect to Pb and (f) Nyquist plots of CsPbBr₃ nanocrystals, Fe(II) doped CsPbBr₃ nanocrystals and with only Fe(II)acetate salt. Inset shows the electrochemical circuit used for calculation.

These as synthesized CsPbBr₃ and Fe(II) doped CsPbBr₃ nanocrystal were further explored for photocatalytic CO₂ reduction in ethyl acetate/water medium²⁰ with an incident light of 150 mW cm⁻² intensity. Ethyl acetate medium was chosen because of its mild polarity and high solubility of CO₂ and this had also been optimized in previous reports.¹⁶ Details of the experimental procedure is provided in supporting information. The photocatalytic performances of undoped, Fe(II) doped CsPbBr₃ and only Fe(II)acetate salt as control reaction are presented in Figure 3. Plots of time dependent evolution of the products CH₄ and CO using these catalysts are provided in Figure 3a and Figure 3b respectively. Importantly, it has been observed that there are differences in product selectivity of the undoped and doped photocatalysts towards CO₂ reduction. The predominated products CO (4.6 μmol g⁻¹ h⁻¹) and CH₄ (1.9 μmol g⁻¹ h⁻¹) were obtained for undoped CsPbBr₃ nanocrystal, which were also shown in previous reports (Table S1).^{17, 19} However, doped nanocrystals predominantly formed CH₄ in comparison to CO. In optimized Fe doping (loading 25% with respect to Pb), the maximum amount of CH₄ (6.1 μmol g⁻¹ h⁻¹) was evolved over CO (3.2 μmol g⁻¹ h⁻¹). Moreover, addition of excess Fe amount was harmful to the photocatalytic performance, resulting in lowering the amount of product formation which was shown in Figure S2. Corresponding energy dispersive spectra (EDS) and elemental compositions are shown in Figure S3. Further, to observe the impact of Fe(II)acetate salt without nanocrystals, the control reaction was also carried out in ethyl acetate/water medium during catalysis; but the reactions were observed slower compared to the presence of nanocrystals catalysts. Other possible reaction by mixing Fe(II)acetate salts with CsPbBr₃ nanocrystals without doping was also measured under similar condition; but, the rate of reduction was observed much slower (Figure S4). Combining all these results including control reaction without Fe(II) and nanocrystals, the evolved CH₄ and CO gases were shown quantitatively in Figure 3c. From all these observations, the catalytic activity for selective production of CH₄ was observed for Fe(II) doped CsPbBr₃

nanocrystals. Furthermore, to confirm the generated CH_4 and CO gases produced from photocatalytic CO_2 reduction reaction, control experiment was also carried out without CO_2 under the same condition. Results (Figure S5) showed insignificant amount of CO which might be from partial photo-oxidation of ethyl acetate or other trace organic contamination; but no CH_4 was traced.

Furthermore, to understand the ability of transportation of photo-generated charge carriers, these nanostructures were explored for photoelectrochemical measurements. These measurements were carried out in 0.1 M solution of tetrabutyl ammonium hexafluorophosphate (TBAPF_6) in ethyl acetate under 150 mW cm^{-2} solar light illumination. Figure 3d represents the chronoamperometric I-t plot at -0.4 V vs Ag/AgCl of the undoped, Fe(II) doped CsPbBr_3 and only Fe(II) acetate salt. From the plot, it was evident that the photocurrent was repeatable with light on-off cycles. The maximum PEC photocurrent density $-120 \mu\text{A cm}^{-2}$ observed for Fe(II) doped CsPbBr_3 (25% Fe loading w.r.t. Pb) nanocrystal which was 3.15 times enhancement than undoped nanocrystals ($38 \mu\text{A cm}^{-2}$). Figure 3e presents the chronoamperometric I-t plots of the photoelectrodes using different percentage of Fe(II) doped CsPbBr_3 nanocrystals and measured at -0.4V vs Ag/AgCl. The photocurrent density observed gradually increased from 0-25% and then decreased with higher doping percentage. This change in the trend was also found in good agreement with photocatalytic CO_2 reduction efficiency (Figure S2). PEC impedance spectra were measured to detect the charge-transport behavior of prepared electrodes and shown in Figure 3f. Nyquist plot suggested the obvious decrease in charge-transfer resistance (R_{ct}) for Fe(II) doped CsPbBr_3 photoelectrode in comparison to undoped and the controlled Fe(II) electrodes. Using the simple R-C circuit model²⁴ the fitted values are provided in Table S2. Hence, all the PEC data were observed in good agreement with the photo-catalytic CO_2 reduction activity and doped nanocrystals were superior.

Further, to observe the sustainability of these nanocrystals, post catalyzed samples were also analyzed. These nanocrystals in ethyl acetate were agglomerated and not properly soluble, and hence, these were precipitated with centrifugations and redispersed in hexane for optical measurements. Figure 4a presents the absorption and PL spectra of post catalyzed CsPbBr_3 and

Fe(II) doped CsPbBr_3 nanocrystals, and all these retained almost the identical spectra. In some cases, scatterings were observed because of agglomeration; but with dilution, clear spectra could be obtained. Figure 4b shows powder X-ray diffraction (XRD) patterns of both doped and undoped post catalyzed nanocrystals and both retained orthorhombic phase. Figure 4c shows the TEM image of post catalyzed CsPbBr_3 and 4d shows that of Fe(II) doped CsPbBr_3 , which retained the cubic shape. In some places on the grid, we observed agglomerations (Figure S6) which were expected because of poor solubility in polar ethyl acetate medium. Energy dispersive spectra (EDS) of Fe(II) doped CsPbBr_3 after 3 hours exposure to light during photocatalysis and further with purification is shown in Figure 4e. The calculated atomic percentage of Fe with 25% (with respect to Pb) loading during the preparation of doped nanocrystals showed 0.76 atomic percentage retained within the nanocrystals after catalysis. The amount retained almost 3% Fe(II) with respect to Pb(II) in the perovskite nanocrystals.

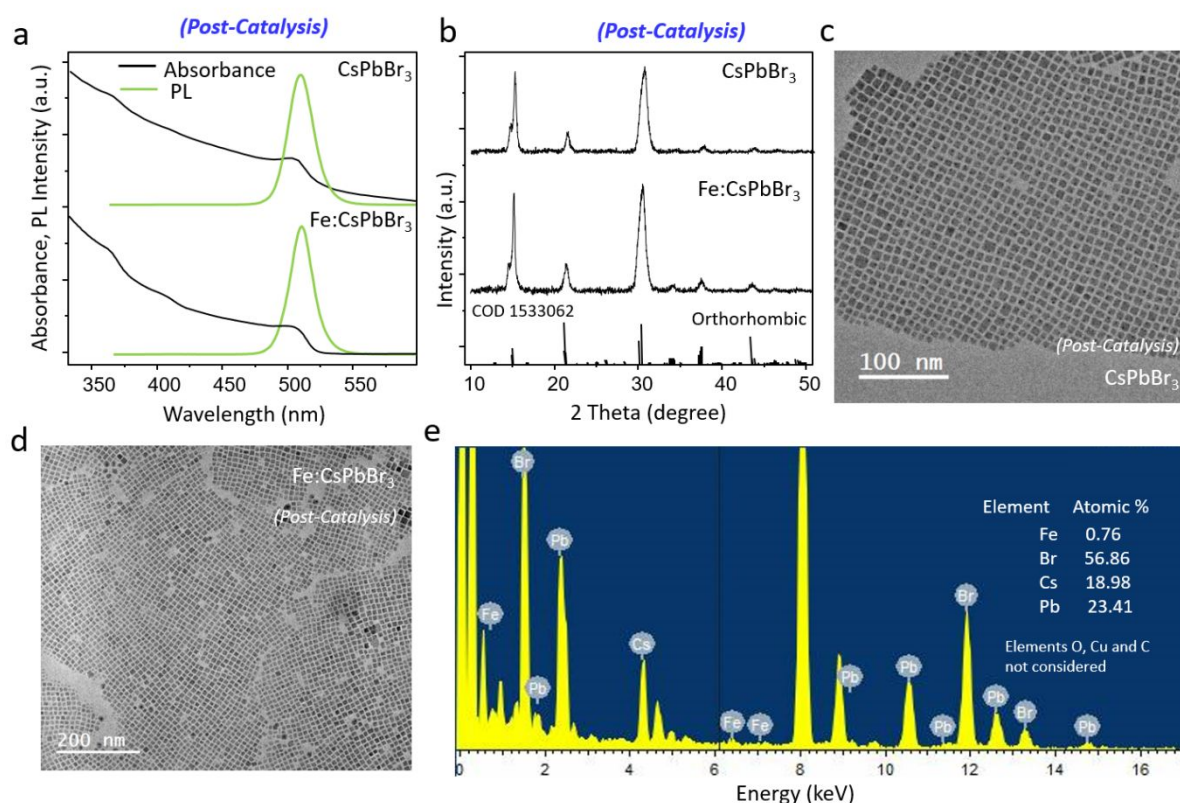


Figure 4. (a) Absorption and PL spectra of post catalyzed CsPbBr_3 and Fe(II) doped CsPbBr_3 nanocrystals. (b) Powder X-ray diffraction patterns of post catalyzed undoped and doped CsPbBr_3

nanocrystals. (c) TEM image of post catalyzed CsPbBr₃ nanocrystals. (d) TEM image of post catalyzed Fe(II) doped CsPbBr₃ nanocrystals. (e) EDS spectra of post catalyzed Fe(II) doped CsPbBr₃ nanocrystals. Excited wavelength for the PL spectra was 350 nm.

From all these above results, it could be confirmed here that Fe(II) doped CsPbBr₃ perovskite nanocrystals acted as better catalysts and also performed selectivity in CO₂ to methane production. Even doping transition metal ions were extensively studied for stabilizing and enhancing photoluminescence intensity, increasing the photoresponse property, obtaining dopant induced new emission etc.,^{4, 25-30} but this was not explored for photocatalytic CO₂ reduction. Hence, it became important for understanding the underlying mechanism of the carrier transportation during photocatalysis. As stated above, Tang et al.²³ theoretically predicted the ability of such reduction using several metal ions. For the case of Fe(II) doping, the adsorption energy for CH₄ was reported more positive indicating CH₄ active molecules once formed were faster desorbed from the catalyst surfaces. This enhanced the reaction in favor of methane production in Fe(II) doped CsPbBr₃ and CO for undoped CsPbBr₃ nanocrystals. This desorption mechanism in energy diagram is shown in Figure 5. However, while we tried the same reactions with Co(II) which was also predicted as efficient dopant; but poor catalytic performance was recorded (Figure S7). This might need modifications of the reaction procedure or a different catalytic strategy.

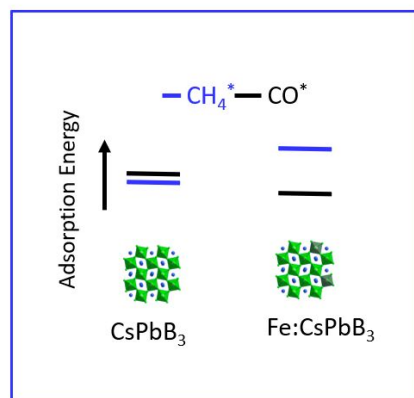


Figure 5. Schematic presentation of the energy diagram of adsorption of active CH₄ and CO molecules onto the surface of undoped and doped CsPbBr₃ nanocrystals.

While the entire study was carried out using Fe(II) ions for doping, nanocrystals were also synthesized using Fe(III) precursors. Unfortunately, efficient doping could not be obtained under similar conditions and Fe could not be detected in purified nanocrystals (Figure S8). Hence, it could be stated here that Fe(II) facilitated the doping forming octahedra like Pb(II) under our adopted reaction protocol.

In conclusion, Fe(II) doped CsPbBr₃ nanocrystals as product selective and efficient photocatalyst for reduction of CO₂ is established. While undoped CsPbBr₃ led to CO as major product, doping Fe(II) helped evolving CH₄ as the predominated gas from photocatalytic reduction of CO₂. Unlike previous reports, these photocatalysts did not require any additional carrier transportation material or heterostructures. Transient current measurement also supported Fe(II) doped nanocrystals as better photoconductor and showed superior photoresponse compared to it undoped counterpart. Carrying out several controlled reactions and analyzing the post catalysis samples, it was established that doping in CsPbBr₃ indeed enhanced the catalytic activities of these nanocrystals. These results with selective CH₄ and CO as major product for doped and undoped nanocrystals were further correlated with their adsorption and desorption abilities onto the catalyst surfaces, which have been predicted theoretically.²³ To our best knowledge these findings of product selectivity is new of its kinds and would help for wider understanding the catalytic activities of these newly emerged perovskite nanomaterials.

Supporting Information: Materials, methods, instrumentations, photo-electrochemical measurement procedure, photochemical CO₂ reduction procedure, EDS spectra, TEM images, gas-chromatography data, chronoamperometry I-t plot, comparison table of literature reports and table of electrochemical circuit parameters.

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